CRANKSHAFT POSITION (CKP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0140q1800500

Visual Inspection

Caution
- **When replacing the CKP sensor, make sure there is no foreign matter on it such as metal shavings. If it is installed with foreign matter, the sensor output signal will malfunction resulting from fluctuation in magnetic flux and cause a deterioration in engine control.**

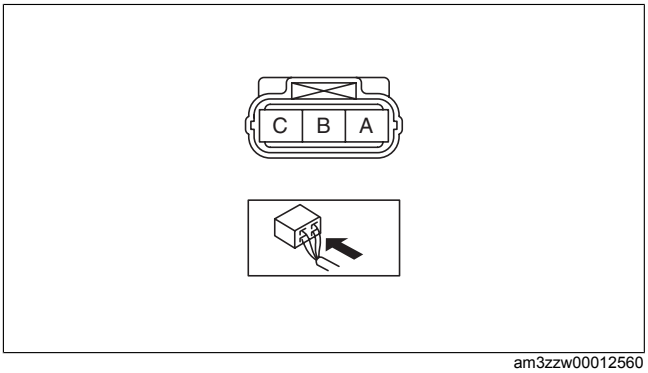
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Lift up the vehicle.
3. Remove the service hole cover (installed to front under cover No.2). (See OIL FILTER REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Verify that there are no metal shavings on the sensor.
   - If there is a malfunction, remove any metal shavings that are adhering.

Voltage Inspection

Caution
- **When replacing the CKP sensor, make sure there is no foreign matter on it such as metal shavings. If it is installed with foreign matter, the sensor output signal will malfunction resulting from fluctuation in magnetic flux and cause a deterioration in engine control.**
- **If the wiring harnesses or waterproof connectors are damaged, water penetrating the connector will cause a sensor malfunction. To prevent this, be careful not to damage wiring harnesses or waterproof connectors.**

1. Idle the engine.
2. Measure the CKP sensor output voltage wave pattern using an oscilloscope.
   - If not as specified, replace the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

- **Specification**
  4.5 V or more (Maximum value of wave pattern)
  0.8 V or less (Minimum value of wave pattern)

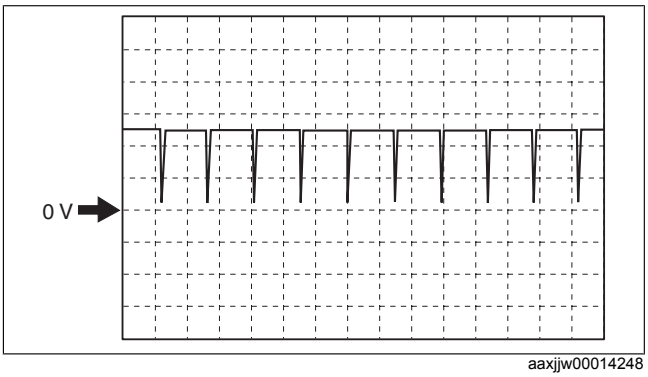


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Wave pattern (reference)

Oscilloscope setting
- **2 V/DIV (Y), 1 ms/DIV (X), DC range**

Vehicle condition
- **Idle (after warm up)**



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